

PACE: A Pedestrian-crossing Automated Classification Engine for Street-Level Images

Gianmaria D'Agostino, Matteo Di Salvo, Andrea Musumeci, Simone Polselli
Department of Electrical and Information Engineering
University of Cassino and Southern Lazio, Cassino, Italy

Abstract—We present PACE (Pedestrian-crossing Automated Classification Engine), a system for the automatic detection of pedestrian crossing traffic signs in street-level images, developed with the OpenCV framework and based exclusively on classical image processing techniques. The proposed pipeline first extracts candidate regions by color-based segmentation and morphological filtering, then verifies each candidate by recognizing the triangular pictogram of the sign through Hu-moment shape matching and geometric constraints. Two alternative strategies for the analysis of the candidate regions are proposed and compared. The system is evaluated on a subset of the Mapillary Traffic Sign dataset consisting of 155 images containing 201 annotated signs, achieving 92.0% precision at 80.1% recall (method 1) and 87.2% precision at 81.1% recall (method 2).

Index Terms—traffic sign detection, image processing, mathematical morphology, shape matching, OpenCV

I. INTRODUCTION

The goal of this work is the detection of pedestrian crossing traffic signs (label information--pedestrians-crossing--g1) in street-level images, using only classical image processing techniques, i.e., without any machine learning component.

The detection problem is challenging for several reasons: the signs appear at very different scales and perspectives, illumination varies considerably across the dataset, and visually similar objects (billboards, other blue signs, windows) act as distractors. The proposed system, PACE (Pedestrian-crossing Automated Classification Engine), addresses the problem with a two-stage pipeline:

- 1) *candidate extraction*: the full image is preprocessed and segmented in order to isolate connected components whose color and shape are compatible with the rectangular plate of the sign;
- 2) *candidate verification*: each candidate region is cropped and further processed in order to detect the triangular pictogram characteristic of the pedestrian crossing sign. Two alternative processing methods are proposed for this stage.

Detections are validated against the ground-truth annotations by means of the Intersection over Union (IoU) criterion, and the overall performance is summarized by precision, recall, and the area under the precision-recall curve (AUPR).

Project developed for the Image Processing & Analysis course (a.y. 2022–2023), under the supervision of Prof. Alessandro Bria.

The remainder of this paper is organized as follows. Section II describes the dataset and the ground-truth format. Section III presents the full-image preprocessing and the candidate extraction criteria. Section IV details the two processing methods applied to the candidate regions, and Section V the criteria used to recognize the triangular pictogram. Section VI defines the evaluation protocol, Section VII reports and discusses the results, and Section VIII draws the conclusions.

II. DATASET AND GROUND TRUTH

The system is evaluated on a subset of the Mapillary Traffic Sign dataset consisting of 155 street-level images, each accompanied by a JSON annotation file. Every annotation lists the objects present in the image together with their class label and bounding box (x_{min} , y_{min} , x_{max} , y_{max}).

At load time, the annotations are parsed and the bounding boxes of all objects labeled information--pedestrians-crossing--g1 are collected into a per-image list of ground-truth rectangles. Overall, the dataset contains 201 annotated pedestrian crossing signs.

III. CANDIDATE EXTRACTION

A. Full-Image Preprocessing

The purpose of the preprocessing stage is to produce a binary mask in which the plates of the traffic signs emerge as compact connected components. The following operations are applied to each image; all kernel sizes and thresholds were tuned empirically on this dataset and are summarized in Table I.

- 1) *Noise reduction*: the original image (Fig. 1) is smoothed with a Gaussian filter with a 17×27 kernel, followed by a median filter with aperture 9. The Gaussian filter suppresses high-frequency noise, while the median filter removes impulsive noise while preserving edges (Fig. 2).
- 2) *Contrast enhancement*: the image is converted to the HSV color space and CLAHE (Contrast Limited Adaptive Histogram Equalization, clip limit 6, 8×8 tiles) is applied to the V channel, improving the local contrast of the luminance (Figs. 3 and 4).
- 3) *Saturation top-hat*: a morphological top-hat transform with a 75×75 rectangular structuring element is applied to the S channel. The top-hat transform highlights compact regions that are more saturated than their



Fig. 1. Original image.



Fig. 3. V channel before CLAHE.



Fig. 2. Image after noise reduction (Gaussian and median filtering).



Fig. 4. V channel after CLAHE.

surroundings, such as the blue plate of the sign (Figs. 5 and 6).

- 4) *Color segmentation*: the processed HSV channels are merged and converted back to BGR (Fig. 7) and then to the Lab color space, where the image is binarized by thresholding each channel: $L \in [20, 220]$, $a \in [40, 140]$, $b \in [50, 118]$ (Fig. 8).
- 5) *Morphological closing*: a closing with a 30×30 rectangular structuring element fills holes and merges fragmented responses belonging to the same object (Fig. 9).

Finally, the external contours of the binary mask are extracted with `findContours` and constitute the set of candidate objects.

B. Candidate Filtering

Each candidate contour is accepted only if it satisfies two criteria:

- 1) *area criterion*: the contour area A must satisfy $A_{\min} \leq A \leq A_{\max}$, with $A_{\min} = 2000$ and $A_{\max} = 100\,000$ pixels; this filter discards components that are too small to be reliably analyzed or too large to correspond to a traffic sign;
- 2) *rectangularity criterion*: the ratio between the contour area and the area of its minimum-area enclosing rectangle (computed with an orientation-corrected `minAreaRect`) must exceed 0.4; this filter retains components whose shape is compatible with the rectangular plate of the sign.

For every accepted candidate, the axis-aligned bounding rectangle is computed and the corresponding region is cropped from the image (Fig. 10); all subsequent processing operates on this crop.



Fig. 5. S channel before the top-hat transform.



Fig. 6. S channel after the top-hat transform (75×75 structuring element).

IV. PROCESSING OF THE CANDIDATE REGIONS

Two alternative methods were developed to binarize the cropped candidate regions so as to isolate the triangular pictogram of the sign. Both methods produce a binary image from which contours are extracted with `findContours`.

A. Method 1: Grayscale Top-Hat

- 1) Before cropping, the contour of the candidate is drawn on the image in blue, i.e., the dominant color of the sign. In the subsequent grayscale conversion this outline becomes a dark border that separates the pictogram from the background of the crop (Fig. 11).
- 2) The crop is converted to HSV and CLAHE (clip limit 6, 14×14 tiles) is applied to the V channel to enhance the local contrast (Fig. 12); the channels are then merged and converted back to BGR.
- 3) A Gaussian filter (3×3) and a median filter (aperture 3) reduce the residual noise (Fig. 13).



Fig. 7. Color image after preprocessing (merged HSV channels converted back to BGR).



Fig. 8. Binary mask obtained by thresholding in the Lab color space.

- 4) The crop is converted to grayscale and a top-hat transform with a 140×140 rectangular structuring element is applied, highlighting the bright pictogram against its darker surroundings (Fig. 14).
- 5) The result is binarized with Otsu's method, which automatically selects the threshold that best separates the two pixel classes, and a morphological closing with a 5×5 structuring element regularizes the resulting shapes (Fig. 15).

B. Method 2: Saturation Top-Hat

- 1) The crop is smoothed with a Gaussian filter (15×15) followed by a median filter (aperture 5) to reduce noise (Fig. 16).
- 2) The crop is converted to HSV and a top-hat transform with a 140×140 rectangular structuring element is



Fig. 9. Binary mask after morphological closing (30×30 structuring element).



Fig. 10. Cropped candidate region.



Fig. 11. Cropped region without (left) and with (right) the candidate contour drawn in blue.



Fig. 12. V channel of the crop before (left) and after (right) CLAHE.



Fig. 13. Crop after channel merging (left) and after Gaussian and median filtering (right).



Fig. 14. Grayscale crop before (left) and after (right) the top-hat transform.



Fig. 15. Otsu binarization before (left) and after (right) the 5×5 closing.

applied to the S channel, enhancing the highly saturated regions of the sign (Fig. 17).

- 3) The S channel is binarized with Otsu's method and regularized with a morphological closing with a 15×15 structuring element (Fig. 18).
- 4) The binary image is inverted with `bitwise_not`, so that the pictogram is represented by white pixels (Fig. 19).

V. SIGN VERIFICATION

The contours extracted from the binarized crop are examined in order to establish whether the candidate region actually contains a pedestrian crossing sign. A candidate contour is accepted as a detection if it satisfies all of the following criteria:

- 1) *area criterion*: the contour area must lie between 500 and 30 000 pixels;
- 2) *shape matching*: the similarity between the contour and a reference triangle template is measured with



Fig. 16. Crop before (left) and after (right) Gaussian and median filtering.

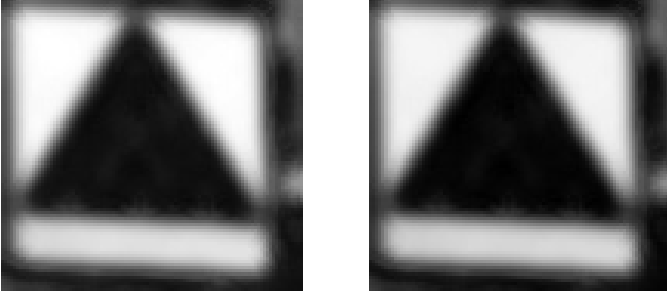


Fig. 17. S channel of the crop before (left) and after (right) the top-hat transform.



Fig. 18. Otsu binarization of the S channel before (left) and after (right) the 15×15 closing.

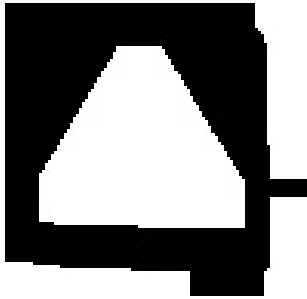


Fig. 19. Inverted binary image: the pictogram is represented by white pixels.

`matchShapes` (Hu-moment based, comparison mode `CONTOURS_MATCH_11`); the match score must not exceed 0.13 for method 1 and 0.2 for method 2, lower scores indicating a closer match;

- 3) *geometric verification*: the contour is simplified with `approxPolyDP` (approximation tolerance $\varepsilon = 40$

TABLE I
SUMMARY OF THE PIPELINE PARAMETERS

Stage	Operation	Parameters
Preprocessing	Gaussian blur	17×27 kernel
	Median blur	aperture 9
	CLAHE (V)	clip 6, 8×8 tiles
	Top-hat (S)	75×75 SE
	Lab thresholds	$L [20, 220]$, $a [40, 140]$, $b [50, 118]$
	Closing	30×30 SE
Cand. filter	Area	$2\,000\text{--}100\,000 \text{ px}^2$
	Rectangularity	> 0.4
Method 1	CLAHE (V)	clip 6, 14×14 tiles
	Gauss. + median	3×3 ; aperture 3
	Top-hat (gray)	140×140 SE
	Otsu + closing	5×5 SE
Method 2	Gauss. + median	15×15 ; aperture 5
	Top-hat (S)	140×140 SE
	Otsu + closing	15×15 SE, inverted
Verification	Area	$500\text{--}30\,000 \text{ px}^2$
	<code>matchShapes</code>	≤ 0.13 (M1); ≤ 0.2 (M2)
	<code>approxPolyDP</code>	$\varepsilon = 40 \text{ px}$
	Angles	$\pm 28^\circ$ of 60°
Evaluation	IoU threshold	0.2

pixels) and accepted only if the resulting polygon (i) has exactly three vertices, (ii) has sides of similar length (pairwise differences below 2ε), and (iii) has internal angles within $\pm 28^\circ$ of 60° , i.e., is approximately equilateral.

When a contour passes all three tests, its bounding rectangle is mapped back to full-image coordinates and reported as a detection.

VI. EVALUATION PROTOCOL

Detections are validated against the ground-truth rectangles by means of the Intersection over Union criterion. Given a detected rectangle D and a ground-truth rectangle G , the IoU is defined as

$$\text{IoU}(D, G) = \frac{|D \cap G|}{|D \cup G|}, \quad (1)$$

where $|\cdot|$ denotes the area. A detection is counted as a *true positive* (TP) if its IoU with at least one ground-truth rectangle exceeds 0.2, and as a *false positive* (FP) otherwise; ground-truth signs that are not matched by any detection are counted as *false negatives* (FN).

The overall performance is summarized by precision and recall:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (2)$$

To characterize the trade-off between the two metrics, each detection is recorded together with a binary label (1 for true positives, 0 for false positives) and its `matchShapes` score. Since lower scores correspond to more confident detections,

TABLE II
DETECTION PERFORMANCE ON THE FULL DATASET

Method	Precision	Recall	AUPR
Method 1 — grayscale top-hat	92.0%	80.1%	0.787
Method 2 — saturation top-hat	87.2%	81.1%	0.794

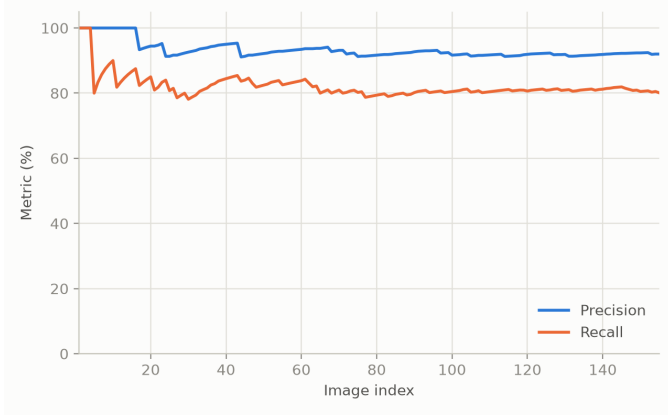


Fig. 20. Method 1: running precision and recall over the dataset.

the detections are ranked by increasing score and the score threshold is swept to trace the precision-recall curve, from which the area under the curve (AUPR) is computed.

VII. RESULTS

Table II reports the performance of the two methods on the full dataset (155 images, 201 annotated signs). The corresponding precision-recall curves are shown in Figs. 21 and 23, and the evolution of the running precision and recall over the dataset in Figs. 20 and 22.

The two methods achieve comparable recall, while method 1 attains a higher precision. The blue outline drawn around the candidate plate (Section IV-A) proves effective in separating the pictogram from the background, reducing the number of spurious triangular responses. Method 2, operating on the saturation channel, is slightly more sensitive but also more prone to false positives on saturated distractors. Most of the missed detections correspond to signs that are small, partially occluded, or poorly contrasted, for which the candidate plate does not survive the segmentation stage.

Since the pipeline parameters were tuned on the same 155 images used for evaluation, Table II measures performance on the development set rather than generalization. To quantify the difference, we additionally evaluated the frozen pipeline on a held-out set of 1,188 MTSD images (1,908 annotated signs) never used during development: precision and recall drop to 65.6%/14.5% for method 1 and 52.5%/13.4% for method 2. The gap is explained partly by scale — 41% of the held-out signs are smaller than the minimum candidate area, whereas the development subset contains none — and partly by the hand-tuned thresholds overfitting the curated subset.

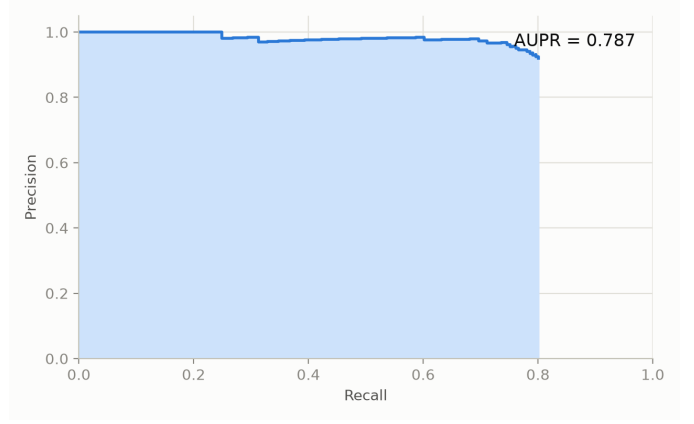


Fig. 21. Method 1: precision-recall curve and AUPR.

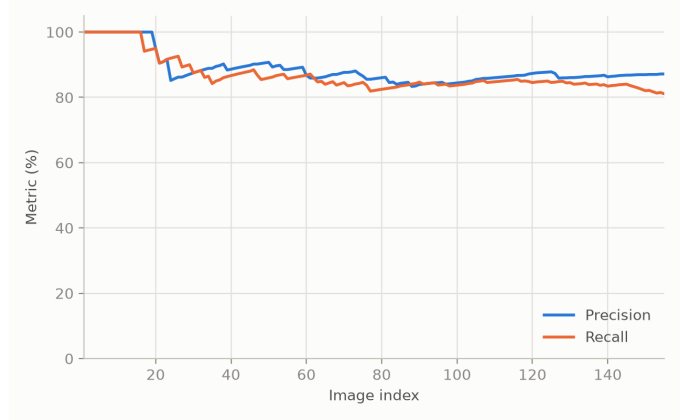


Fig. 22. Method 2: running precision and recall over the dataset.

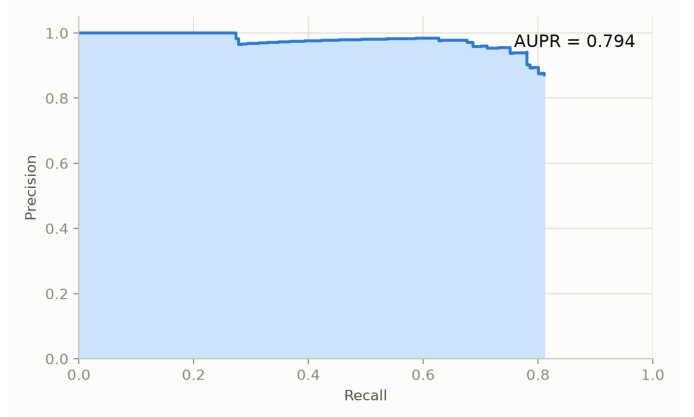


Fig. 23. Method 2: precision-recall curve and AUPR.

VIII. CONCLUSIONS

We presented a classical image processing pipeline for the detection of pedestrian crossing traffic signs in street-level images. The combination of color-based candidate extraction and shape-based verification proves effective on the considered dataset, reaching a precision of 92% at a recall of 80% without any learning component. The held-out evaluation, however,

shows that a hand-tuned classical pipeline does not generalize to the full diversity of street-level imagery, which is the expected ceiling of this didactic exercise. Future work could address the main failure cases — small and low-contrast signs — by introducing multi-scale processing or adaptive segmentation thresholds; more fundamentally, given the size of the full MTSD dataset (over 40,000 annotated images), the natural continuation of this work is a learning-based detector, with the present pipeline serving as an interpretable baseline.